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Paper

A closed-form reference distribution for goodness-of-fit testing under penalised logistic regression in the proportional regime

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Abstract

Grouped goodness-of-fit tests for logistic regression assume maximum likelihood fitting. Under penalised regression, and ridge regression in particular, shrinkage displaces the grouped residuals; a recent correction removes that displacement but obtains a valid reference only by a bootstrap with refits. The analytic first-order reference fails because it standardises the residuals by the fitted Bernoulli variance, which shrinkage inflates towards one quarter, an error reaching 19 per cent when predictors are a quarter of the sample. Evaluating the variance at the debiased fit makes the reference asymptotically exact at fixed dimension. In the proportional regime of high-dimensional logistic regression we estimate the null variance from one fit by the observable adjustments of regularised M-estimation, and identify the remaining term as self-selection by grouping on a noisy index, concentrated in the cubic direction. The resulting test of calibration along the fitted index, CALM, needs no resampling and matches the bootstrap in power. On a smooth basis whose degree is chosen from an observable measure of index noise it held its level in every design tried, for aspect ratios up to 0.4; the decile basis needs a calibrated correction that does not transfer everywhere. A glaucoma analysis runs in under a second.

Keywords calibration; goodness-of-fit; high-dimensional logistic regression; penalised regression; proportional regime; ridge regression

1. Introduction

Penalised logistic regression is now the recommended way to build a clinical prediction model when the number of candidate predictors is large relative to the number of events (Riley et al., 2021; Pavlou et al., 2024), and the fitted model is then checked for calibration in the usual way, almost always with a grouped goodness-of-fit test such as the Hosmer–Lemeshow test (Hosmer and Lemeshow, 1980; Hosmer et al., 1997). Ebrahim (2026) showed that this combination is not valid: under a ridge penalty the grouped standardised residuals acquire a non-centrality induced by shrinkage, and at the penalty that cross-validation selects the test rejects correctly specified models between 92 and 100 per cent of the time. The same paper repaired the test by subtracting an estimate of the shrinkage displacement, which restores the ordinary maximum likelihood covariance to first order, and then obtained a valid procedure by prepivoting (Beran, 1987, 1988) — a bootstrap in which the penalised model is refitted in every resample. The correction was necessary; the bootstrap was needed because the first-order theory did not deliver a usable reference distribution on its own. The analytic reference was conservative, and increasingly so as the ratio of predictors to observations grew.

This paper explains that failure and removes it. The explanation is simple once seen. The grouped residuals are standardised by the Bernoulli variance $\hat{\pi}_i(1-\hat{\pi}_i)$ evaluated at the *fitted* probabilities. Ridge regression pulls every fitted probability towards one half, where $\pi(1-\pi)$ is largest, so the standardisation uses a variance that is too large and every residual looks smaller than it is. The error is of order λ/n, where λ is the penalty and n the sample size; it is negligible in the asymptotic regime of the first-order theory and material at the penalties analysts actually use — 7 per cent of the statistic at $\lambda/n = 0.2$ and 19 per cent when the number of predictors is a quarter of the sample size. Evaluating the variance at the debiased fit instead makes the analytic reference exact wherever the first-order theory applies.

In the proportional regime, where $p/n \to \kappa \in (0, 1)$, two further things happen, and both are dealt with here. First, the debiased fit is itself noisy, so its Bernoulli variance understates the truth; we use the observable adjustments of Bellec (2025) for regularised M-estimation to estimate the null variance from the fit alone. Second, grouping on a noisy fitted index selects observations by their own noise, which leaves a non-centrality in the corrected residual. We show that this term is of order κ, that it is concentrated almost entirely in the cubic direction of the group index, and that it is removed by a basis whose polynomial degree is chosen from an observable measure of index noise — a rule that the attenuation law of Ebrahim (2026) already recommends on grounds of power. The result is a test with a closed-form reference distribution and no resampling. We call it CALM, for calibration assessment under lambda-shrunk models. Tests for high-dimensional regression built by other routes — the residual prediction tests of Shah and Bühlmann (2018) for linear models and Janková et al. (2020) for generalised linear models, and the coefficient tests of Guo and Chen (2016) — ask whether the model is correct as a function of the covariates, or whether a block of coefficients is zero; the statistics treated here ask whether the deployed predictor is calibrated along its own index, and Ebrahim (2026) showed that the two families reject on essentially disjoint datasets. The present paper supplies the analytic reference that the second family lacked.

The contributions are as follows. (a) Theorem 1 identifies the fitted-variance error and gives the corrected first-order law; the lesson extends to any residual diagnostic for a penalised generalised linear model that standardises by the fitted variance. (b) Proposition 1 gives an observable estimate of the null variance in the proportional regime, and Section 4.3 characterises the selection non-centrality, its order in κ and its cubic structure. (c) Corollary 1 turns the attenuation law of Ebrahim (2026), restated here as Proposition 2, into an adaptive basis that retains the directions in which a departure survives grouping and, as a consequence, removes the selection term. (d) Section 5 separates the two bases. On the adaptive EDGE basis, which by construction avoids the direction the selection term occupies, CALM held its level in all twenty-five designs we examined, for κ up to 0.4, for three predictor laws and for designs chosen after the constants were fixed. On the decile basis it needs the calibrated correction, and there we report the designs in which that correction is too small. In power it matches prepivoting, and Section 5.5 compares it with the high-dimensional test of Janková et al. (2020) and with the calibration checks in clinical use, which do not hold their level on a penalised fit. Section 6 returns to the glaucoma data of Ebrahim (2026), and Section 7 states what is proved, what is calibrated, and where the boundary lies.

2. The shrinkage-corrected statistics and their first-order law

2.1. Notation

Let $y_i \in \{0, 1\}$ and $\boldsymbol{x}_i \in \mathbb{R}^p$, $i = 1, \ldots, n$, and write X for the $n \times (p+1)$ design matrix including the intercept column. The logistic model specifies $\pi_i(\boldsymbol{\beta}) = \{1 + \exp(-\boldsymbol{x}_i^\top \boldsymbol{\beta})\}^{-1}$, and penalised estimation solves $\hat{\boldsymbol{\beta}} = \arg\max_{\boldsymbol{\beta}} \{\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda \boldsymbol{\beta}^\top D \boldsymbol{\beta}\}$ with $D = \mathrm{diag}(0, 1, \ldots, 1)$, the intercept unpenalised (le Cessie and van Houwelingen, 1992). The penalty is on the scale of the log-likelihood; the implementation of Friedman et al. (2010) minimises $-n^{-1}\ell(\boldsymbol{\beta}) + \frac{1}{2}\lambda_{\mathrm{g}} \|\boldsymbol{\beta}_{-0}\|^2$, so $\lambda = n\lambda_{\mathrm{g}}$, and we report both scales. Write $\hat{\pi}_i = \pi_i(\hat{\boldsymbol{\beta}})$, $W = \mathrm{diag}\{\hat{\pi}_i(1 - \hat{\pi}_i)\}$, $F = X^\top W X$, $K = \lambda D$ and $M = F + K$. Let C be the $G \times n$ indicator matrix of the G equal-frequency groups of the fitted probabilities, $V = CWC^\top$, and

$$\boldsymbol{r} = V^{-1/2} C(\boldsymbol{y} - \hat{\boldsymbol{\pi}}), \qquad U = V^{-1/2} CWX. \tag{1}$$

The Hosmer–Lemeshow statistic is $\|\boldsymbol{r}\|^2$ up to the choice of denominator. Two bases are used throughout: the *decile* basis spends all G directions, and the *EDGE* basis of Ebrahim and El-Kotory (2026) projects $\boldsymbol{r}$ onto the orthogonal polynomials of degree one to k in the group-mean fitted probability, Z being the resulting $G \times k$ matrix, with $k = 3$ as the default. We write $P_Z = Z(Z^\top Z)^{-1} Z^\top$.

2.2. The corrected statistics

Under a penalty, $\mathbb{E}(\boldsymbol{r}) \neq \mathbf{0}$ even when the model is correct, because the fitted probabilities are shrunk. Ebrahim (2026) derived the displacement $\boldsymbol{\mu}_K = UM^{-1}K\boldsymbol{\beta}$ to first order and showed that with the one-step debiased estimate $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} + F^{-1}K\hat{\boldsymbol{\beta}}$ and the plug-in $\hat{\boldsymbol{\mu}} = UM^{-1}K\tilde{\boldsymbol{\beta}}$, the corrected residual

$$\boldsymbol{v} = \boldsymbol{r} - \hat{\boldsymbol{\mu}} \tag{2}$$

satisfies, to first order and under regularity conditions with $\lambda = o(n)$ and p fixed,

$$\boldsymbol{v} = A^*(\boldsymbol{y} - \boldsymbol{\pi}) + o_p(1), \qquad A^* = V^{-1/2}C - UF^{-1}X^\top, \tag{3}$$

where $\boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\beta})$ is the true probability vector. The shrinkage-corrected statistics are

$$\mathrm{SC.HL} = \|\boldsymbol{v}\|^2, \qquad \mathrm{SC.EDGE} = \boldsymbol{v}^\top P_Z \boldsymbol{v}. \tag{4}$$

Equation (3) is Proposition 2 of Ebrahim (2026), proved there under Assumption 1 below, which we restate so that this paper is self-contained. We use it as given; the argument of the present paper begins where that one stops, with the covariance of the right-hand side.

Assumption 1 The true probabilities satisfy $\pi_i(\boldsymbol{\beta}) \in [\epsilon, 1-\epsilon]$ for some $\epsilon > 0$ uniformly in i; $n^{-1}F \to \Sigma$ positive definite, and λ is non-random with $\lambda = o(n)$; the number of groups G is fixed and each group receives a fixed positive proportion of the observations; and p is fixed as $n \to \infty$.

The last two conditions are the ones that matter for what follows. The fixed grouping is an assumption rather than a consequence, as Ebrahim (2026) notes; and p fixed with $\lambda = o(n)$ places (3) squarely outside the regime of Section 4, where p/n tends to a positive constant and the penalties used have λ/n of order one. Sections 4.2 to 4.5 therefore build on an expansion whose validity in the proportional regime we assume rather than prove, and check empirically in Section 5. We return to this in Section 7.

The null hypothesis must be stated with care, because two hypotheses that coincide under maximum likelihood separate under a penalty (Ebrahim, 2026). The structural null H_0^{str} says that the logistic form is correct: $\mathbb{E}(y_i \mid \boldsymbol{x}_i) = \pi_i(\boldsymbol{\beta})$ for some $\boldsymbol{\beta}$. The calibration null H_0^{cal} says that the deployed probabilities are calibrated: $\mathbb{E}(y_i \mid \hat{\pi}_i) = \hat{\pi}_i$. Under a penalty a correct model delivers shrunk probabilities, so H_0^{str} can hold while H_0^{cal} fails by an amount the analyst chose with λ (van Houwelingen and le Cessie, 1990). The statistics of (4), and CALM, test H_0^{str} along the fitted index: a non-rejection says the form is right along that index, not that the shrunk probabilities can be used without recalibration.

2.3. Exact tail probabilities

If $\boldsymbol{v} \sim \mathcal{N}(\mathbf{0}, \Sigma)$ then $\|\boldsymbol{v}\|^2$ is distributed as $\sum_l \omega_l \chi^2_{1,l}$ with ω_l the eigenvalues of Σ, and $\boldsymbol{v}^\top P_Z \boldsymbol{v}$ as $\sum_l \nu_l \chi^2_{1,l}$ with ν_l the eigenvalues of $\Sigma^{1/2} P_Z \Sigma^{1/2}$, of which at most k are non-zero. Tail probabilities of such weighted sums are available to any accuracy by the algorithm of Davies (1980), with Imhof (1961) as a fallback; we use the implementation of Duchesne and Lafaye de Micheaux (2010). No chi-squared approximation with a nominal number of degrees of freedom is used anywhere in this paper, and nothing that follows depends on one.

3. Why the first-order reference fails: the fitted variance

3.1. The corrected covariance

The first-order theory refers $\boldsymbol{v}$ to $\mathcal{N}(\mathbf{0}, \Omega_{\mathrm{MLE}})$ with $\Omega_{\mathrm{MLE}} = I_G - UF^{-1}U^\top$, the ordinary maximum likelihood covariance of the grouped residuals, and this reference was found to be conservative in the proportional regime (Ebrahim, 2026, Table S1). The following result says exactly what is wrong with it.

Theorem 1 *Let $W_0 = \operatorname{diag}\{\pi_i(1-\pi_i)\}$ be the Bernoulli variance at the true probabilities, and A^* as in (3). Under the conditions of (3),*

$$\operatorname{var}(\boldsymbol{v}) = A^* W_0 A^{*\top} + o(1). \tag{5}$$

Moreover the identity $A^* W A^{*\top} = I_G - UF^{-1}U^\top = \Omega_{\mathrm{MLE}}$ holds exactly for the fitted W, and

$$\|A^* W_0 A^{*\top} - \Omega_{\mathrm{MLE}}\| = O_p\big(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|\big) = O_p(\lambda/n + n^{-1/2}). \tag{6}$$

Proof By (3), $\operatorname{var}(\boldsymbol{v}) = A^* \operatorname{var}(\boldsymbol{y} - \boldsymbol{\pi}) A^{*\top} + o(1)$, and $\operatorname{var}(y_i - \pi_i) = \pi_i(1-\pi_i)$ because the y_i are independent Bernoulli variables with the true probabilities; this is (5). For the identity, expand $A^* W A^{*\top}$ using the three relations $V^{-1/2} C W C^\top V^{-1/2} = I_G$, $V^{-1/2} C W X = U$ and $X^\top W X = F$, all of which hold because V, U and F are built from the same W:

$$A^* W A^{*\top} = I_G - UF^{-1}U^\top - UF^{-1}U^\top + UF^{-1}FF^{-1}U^\top = I_G - UF^{-1}U^\top.$$

For (6), $A^*(W_0 - W)A^{*\top}$ is bounded by $\|A^*\|^2 \max_i |\pi_i(1-\pi_i) - \hat{\pi}_i(1-\hat{\pi}_i)|$; the map $\pi \mapsto \pi(1-\pi)$ is Lipschitz with constant one, $|\hat{\pi}_i - \pi_i| \le \frac{1}{4} |\boldsymbol{x}_i^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})|$, and $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| = O_p(\lambda/n + n^{-1/2})$ by the expansion in Proposition 1 of Ebrahim (2026), the λ/n term being the shrinkage displacement $M^{-1}K\boldsymbol{\beta}$ and the $n^{-1/2}$ term the estimation error. □

The point of the theorem is the direction of the error. The first-order reference is not wrong in its algebra: Ω_{MLE} is exactly $A^* W A^{*\top}$. The error is in W. Because ridge regression shrinks every $\hat{\pi}_i$ towards one half and $\pi(1-\pi)$ is maximised there, $W \succeq W_0$ in the cases that matter, the standardisation in (1) divides by too large a variance, and $\|\boldsymbol{v}\|^2$ is systematically smaller than $\operatorname{tr}(\Omega_{\mathrm{MLE}})$. In design B of Section 5, at $\kappa = 0.25$ and $\lambda = 416$, the observed mean of SC.HL under a correct model is 4.87 against $\operatorname{tr}(\Omega_{\mathrm{MLE}}) = 6.02$, a deficit of 19 per cent; the same ratio is 0.93 at fixed $p = 5$ with $\lambda/n = 0.2$. Figure 1 shows the deficit in all eight cells examined. The bootstrap of Ebrahim (2026) was never affected, because it generates from the debiased fit and so carries the correct variance implicitly; the analytic reference was.

3.2. The variance at the debiased fit

Theorem 1 suggests the obvious repair: replace W by an estimate of W_0. The natural one is $\widetilde{W} = \operatorname{diag}\{\tilde{\pi}_i(1-\tilde{\pi}_i)\}$ with $\tilde{\pi}_i = \pi_i(\tilde{\boldsymbol{\beta}})$, the Bernoulli variance at the debiased fit, and the reference

$$\Sigma_{\mathrm{d}} = A^* \widetilde{W} A^{*\top}. \tag{7}$$

This is exactly the choice the prepivoting bootstrap makes implicitly, since it generates resampled responses from $\boldsymbol{\pi}(\tilde{\boldsymbol{\beta}})$. Table 1 reports the consequence. At fixed p the deflation ratio moves from 0.93 to 1.00–1.01 and the size from 0.02–0.03 to 0.04–0.05; at $\kappa = 0.10$ from 0.79 to 0.99 and 0.006 to 0.054. Wherever the first-order theory applies, replacing the variance is the whole correction.

At $\kappa \ge 0.25$, however, (7) over-corrects: the deflation ratio rises above one and the size to 0.07–0.13, most where the penalty is lightest. The reason is that $\tilde{\boldsymbol{\beta}}$ is now a noisy estimate — at $\kappa = 0.25$ its correlation with the true index is about 0.7 — so $\tilde{\pi}_i$ is *more* extreme than π_i and $\widetilde{W}$ understates the null variance. Two things are needed in the proportional regime: a variance estimate that does not inherit the noise of $\tilde{\boldsymbol{\beta}}$, and an account of a second term that the proportional regime introduces. The next section supplies both.

4. An observable reference in the proportional regime

4.1. Observable adjustments

Bellec (2025) studies regularised M-estimators $\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{b}} n^{-1} \sum_i \ell_{y_i}(\boldsymbol{x}_i^\top \boldsymbol{b}) + g(\boldsymbol{b})$ in single-index models $y_i = \mathcal{F}(\boldsymbol{x}_i^\top \boldsymbol{w}, \varepsilon_i)$ with Gaussian design, in the regime where p/n is bounded, and shows that the mean-field description of $\hat{\boldsymbol{\beta}}$ and of the fitted values $X\hat{\boldsymbol{\beta}}$ obtained by Sur and Candès (2019), Salehi et al. (2019) and others can be expressed through quantities computed from the data alone. We need three of them, written in our notation. Let $\boldsymbol{\psi} = \boldsymbol{y} - \hat{\boldsymbol{\pi}}$

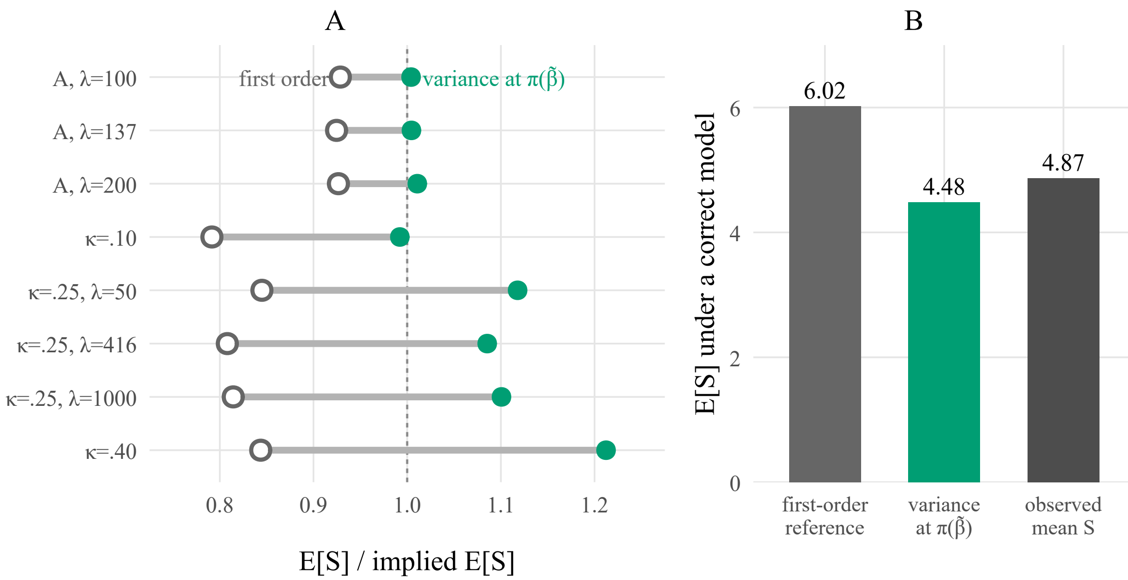

Figure 1 Deflation of the corrected statistic under two references. A: ratio of the observed mean of SC.HL to its implied mean under the first-order reference (open circles) and under the reference with the variance evaluated at the debiased fit (filled circles), eight cells, correct model. B: the three quantities at $\kappa = 0.25$, $\lambda = 416$.

be the raw residual, let $\hat{\mathrm{df}} = \mathrm{tr}(XM^{-1}X^\top W)$, let $\Gamma = W - WXM^{-1}X^\top W$ and $\hat\gamma = \hat{\mathrm{df}}/\,\mathrm{tr}(\Gamma)$; and let $\hat a^2$ and $\hat\sigma^2$ be the estimators of Bellec's equation (3.20) of, respectively, the squared inner product of $\hat{\boldsymbol\beta}$ with the index direction and the squared norm of its component orthogonal to the index. Then Theorem 4.3 of Bellec (2025) gives, for the logistic loss and each observation,

$$\boldsymbol{x}_i^\top \hat{\boldsymbol\beta} - \hat\gamma\psi_i \;\approx\; a_* U_i + \sigma_* Z_i, \tag{8}$$

where $U_i = \boldsymbol{x}_i^\top \boldsymbol{w}$ is the true index standardised to unit variance, Z_i is standard normal and independent of U_i, and (a_*, σ_*) are consistently estimated by $(\hat a, \hat\sigma)$ (his Theorem 4.4). Equation (8) says that an observable quantity, $m_i = \boldsymbol{x}_i^\top \hat{\boldsymbol\beta} - \hat\gamma\psi_i$, equals the true index up to scale plus independent Gaussian noise of known variance. Two points of implementation must be stated. Bellec's model has no intercept and his penalty is strongly convex in every direction, while ours leaves the intercept unpenalised; we centre the design and the fitted index before applying (8) and refit the intercept in (10) below. And his estimator of $\hat a^2$ involves $\Sigma_x^{-1/2}$ with Σ_x known; we replace Σ_x by the sample covariance of the centred design, a plug-in whose error at p/n up to 0.4 is not covered by his Theorem 4.4. Its size can be measured, since in a simulation the true Σ_x is known: over 200 replications the plug-in raises $\hat\rho$ from the oracle value by 0.030 at $\kappa = 0.25$, 0.042 at $\kappa = 0.40$ and 0.034 in the design with weak covariate correlation. The bias is small but it is one-sided, and it is in the direction that makes the rule of Corollary 1 keep the cubic direction slightly too often, which the threshold τ has to absorb.

4.2. The de-noised null variance

Given (8) and a Gaussian design, the conditional law of the true index given the fit is Gaussian:

$$a_* U_i \mid m_i \;\sim\; \mathcal{N}\Big(\frac{\hat a^2}{\hat a^2 + \hat\sigma^2}\, m_i,\; \frac{\hat a^2 \hat\sigma^2}{\hat a^2 + \hat\sigma^2}\Big). \tag{9}$$

The true linear predictor is $\eta_{0i} = \alpha + c\,a_* U_i$ for an intercept α and a scale c that the single-index model leaves free but the known logistic link identifies. We estimate (α, c) by maximising the marginal likelihood of $\boldsymbol{y}$ given the conditional law (9), a one-dimensional integral per observation evaluated by twenty-point Gauss–Hermite quadrature, and define

$$\hat{w}_{0i} = \mathbb{E}\big\{\mathrm{expit}(\hat{\alpha} + \hat{c}\,\zeta)\,[1 - \mathrm{expit}(\hat{\alpha} + \hat{c}\,\zeta)] \;\big|\; m_i\big\}, \qquad \text{where } \zeta \text{ has the conditional law (9)}, \tag{10}$$

the de-noised estimate of the null Bernoulli variance, with $\widehat{W}_0 = \mathrm{diag}(\hat{w}_{0i})$ and the reference $\Sigma_{\mathrm{dn}} = A^*\widehat{W}_0 A^{*\top}$.

Assumption 2 The design has independent rows $\boldsymbol{x}_i \sim \mathcal{N}(\boldsymbol{0}, \Sigma_x)$ with the condition number of Σ_x bounded; p/n is bounded above and, for the rates quoted from Bellec (2025), bounded away from zero, the fixed-dimension case being covered instead by Theorem 1; the penalty satisfies Assumptions A and B of Bellec (2025) after the intercept is removed by centring; the rows of A^* have bounded Euclidean norm; and the logistic scale c is identified and its maximum-likelihood estimate consistent.

Proposition 1 *Under Assumption 2 and a correctly specified logistic model,* $n^{-1}\sum_i |\hat{w}_{0i} - \mathbb{E}\{\pi_i(1-\pi_i) \mid m_i\}| \to 0$ *in probability, and hence* $\Sigma_{\mathrm{dn}} - A^*\overline{W}_0 A^{*\top} \to 0$*, where* $\overline{W}_0 = \mathrm{diag}[\mathbb{E}\{\pi_i(1-\pi_i) \mid m_i\}]$ *is the conditional null variance given the fit.*

Proof By Theorems 4.3 and 4.4 of Bellec (2025), $(\hat{\gamma}, \hat{a}^2, \hat{\sigma}^2)$ converge to $(\gamma_*, a_*^2, \sigma_*^2)$ and the representation (8) holds with an averaged Wasserstein error of order $n^{-1/2}$. With a Gaussian design, (9) is the exact conditional law of $a_* U_i$ given m_i under the limiting representation, so the right-hand side of (10), a bounded Lipschitz functional of $(m_i, \hat{a}, \hat{\sigma}, \hat{\alpha}, \hat{c})$, converges to $\mathbb{E}\{\pi_i(1-\pi_i) \mid m_i\}$ by the continuous-mapping theorem once $(\hat{\alpha}, \hat{c})$ are consistent, which is the last part of the assumption. The statement about Σ_{dn} follows from the row-norm condition in Assumption 2: writing $\boldsymbol{a}_g$ for the gth row of A^* and $d_i = \hat{w}_{0i} - \mathbb{E}\{\pi_i(1-\pi_i) \mid m_i\}$, the (g,h) entry of $A^*\mathrm{diag}(d)A^{*\top}$ is bounded by $\max_g \|\boldsymbol{a}_g\|_\infty^2 \sum_i |d_i|$, which is $O(n^{-1}\sum_i |d_i|)$ because each $\boldsymbol{a}_g$ has entries of order $n^{-1/2}$, and the proposition bounds $n^{-1}\sum_i |d_i|$. $\square$

Three remarks on the proposition. It is stated for the conditional variance given the fit, which is what a reference conditional on the observed grouping requires, and its target is that conditional variance under the limiting representation (8) rather than the exact finite-sample conditional variance; the two differ by the Wasserstein error of Bellec's theorem. And its last hypothesis is not proved here: the scale c enters only through the known link, and consistency of its one-dimensional maximum-likelihood estimate is what one expects but is stated as an assumption so that the reader knows where the proof stops. Estimating the signal strength of a high-dimensional logistic model from data alone is a solved problem for the maximum likelihood estimator, by the ProbeFrontier method of Sur and Candès (2019) and by SLOE (Yadlowsky et al., 2021); those estimators are built for the maximum likelihood fit and do not apply to a ridge fit at a penalty of the analyst's choosing, which is why the scale is estimated here from the conditional law (9). The estimate is constrained to $c \in [0.05, 20]$, a box introduced after an unconstrained search diverged on a small number of replications. Empirically the estimate is good where the penalty is moderate and poor where it is light. Across the twenty-five designs of Section 5 it recovers the true standard deviation of the linear predictor within 3 per cent at fixed p and within 16 per cent in the proportional-regime cells at $\lambda \geq 416$, under Gaussian, heavy-tailed and binary predictors alike; at the lightest penalties it overstates the scale badly, by 21 per cent at $\kappa = 0.25$, $\lambda = 50$ and by 58 per cent at $\kappa = 0.40$, $\lambda = 50$. Those are also the cells in which the decile statistic is least reliable, and the two facts are connected: an overstated scale understates the null variance.

Table 1 shows what the de-noised variance buys. It coincides with (7) at fixed p, repairs the cell in which $\tilde{\boldsymbol{\beta}}$ overshoots most ($\kappa = 0.25$, $\lambda = 50$: size 0.046 against 0.086), and halves the excess at $\kappa = 0.40$. A residual of one to three percentage points remains at $\kappa \geq 0.25$, and it is not a variance.

4.3. The selection non-centrality

The remaining term is a mean. Under a correct model the corrected residual should average zero in every group; at $\kappa = 0.25$ it does not. Over 300 replications of design B at $\lambda = 416$ the squared norm of the mean of $\boldsymbol{v}$ is 0.288 (300 replications; the Monte Carlo floor, the value expected from sampling error alone, is about 0.012), and its shape across the ten groups is the S-curve of Figure 2A. Holding the grouping fixed — forming the deciles from the fitted index of the noise-free response $\boldsymbol{\pi}(\boldsymbol{\beta})$, a function of X alone — reduces it to 0.077, with the deflation ratio unchanged. Most of the term is therefore created by grouping on the fitted index. The remainder is not: 0.077

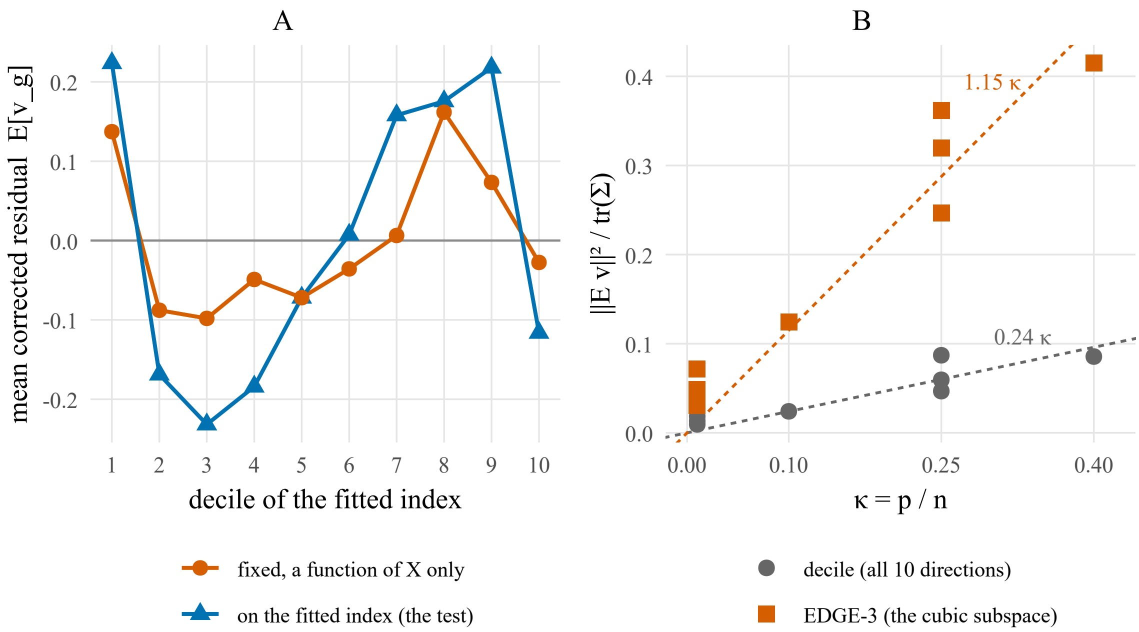

Figure 2 The selection non-centrality. A: mean corrected residual by decile at $\kappa = 0.25$, $\lambda = 416$, when the groups are formed on the fitted index and when they are held fixed as a function of the design. B: the non-centrality fraction against κ for the decile basis and for the three-dimensional EDGE subspace, with the fitted lines 0.24κ and 1.15κ.

is about six times the Monte Carlo floor, and it is the second-order part of the shrinkage displacement, which the plug-in $\hat{\boldsymbol{\mu}}$ removes only to first order. A delta-method expansion of $\hat{\boldsymbol{\mu}}$ about $\tilde{\boldsymbol{\beta}}$ predicts 0.067 for it, close to what is observed, and its effect on the size of the test is at most 0.007 in this cell.

The mechanism is regression to the mean. The fitted index is a noisy version of the true one, and an observation reaches the top decile partly because its own noise was positive; noise is positive when y_i exceeded its expectation, so the top decile is enriched with events beyond what its probabilities justify, and the bottom decile depleted. At fixed p the self-influence of y_i on $\hat{\eta}_i$ (Pregibon, 1981) is $h_{ii} = w_i \boldsymbol{x}_i^\top M^{-1} \boldsymbol{x}_i$, of order p/n, and the effect is negligible; in the proportional regime it is of order κ. This is the random-cells effect of Moore and Spruill (1975), which is $o(1)$ in their setting and $O(1)$ here.

Empirical law 1 *Let $\bar{h} = \hat{\mathrm{df}}/n$ be the mean self-influence. Across the eight cells of Section 5, the fraction $\hat{g} = \|\mathbb{E}\boldsymbol{v}\|^2/\operatorname{tr}(\Sigma_{\mathrm{dn}})$ satisfies $\hat{g} \approx 0.24\,\kappa$ (coefficient of determination 0.93), and equivalently $\hat{g} \approx 0.28\,\bar{h}^{1/2}$; restricted to the three-dimensional EDGE subspace the fraction is $\hat{g}_{\mathrm{E}} \approx 1.15\,\kappa$ (0.97), and 90 to 98 per cent of $\|\mathbb{E}\boldsymbol{v}\|^2$ lies in that subspace when $\kappa > 0$, almost none of it in the directions of degree at most two.*

Empirical law 1 is a fitted relation, not a theorem, and it is set apart from the proved results for that reason; its constants are calibrated on the eight cells of Table 1, not derived, and Section 5.4 reports what happens in cells that were not used to fit them. Its content is in Figure 2B. Two consequences are immediate. First, the non-centrality can be placed in the distribution: a reference $(1 + \hat{g})\Sigma_{\mathrm{dn}}$ has the right mean, and because a scaled central weighted chi-squared with the correct mean has a heavier upper tail than the true non-central one, the adjustment is expected to err towards conservatism, as Table 1 confirms. Second, and more usefully, the term is a cubic. It lives in exactly the direction of the EDGE basis that the attenuation law says carries no power when the index is noisy.

4.4. The adaptive basis

Proposition 2 (Attenuation law, Ebrahim, 2026) *Let η^* and $\hat{\eta}^*$ be the true and fitted linear predictors, standardised, jointly Gaussian with correlation ρ. A departure equal to the degree-k Hermite component of the true predictor survives grouping on the fitted predictor with amplitude ρ^k, and its non-centrality is attenuated by ρ^{2k}.*

The correlation ρ is not a nuisance one must guess. Under (8) it is

$$\hat{\rho} = \frac{\hat{a}}{\sqrt{\hat{a}^2 + \hat{\sigma}^2}}, \tag{11}$$

an observable. It equals 0.94 in design A, 0.71 at $\kappa = 0.10$, 0.69 at $\kappa = 0.25$ — matching the value 0.72 that Ebrahim (2026) measured against the truth — and 0.92 on the glaucoma data. Two approximations are involved in using it. It is the correlation of the adjusted index m_i with the truth, while the groups are formed on $\hat{\eta}_i$, which differs from m_i by the self-influence term of Section 4.3; and Proposition 2 concerns Hermite components of a continuous predictor, while the EDGE basis is a set of orthogonal polynomials over G discrete group means, so "zero otherwise" in the corollary holds up to the error of that discretisation. Both approximations improve as G grows and as the self-influence falls.

Corollary 1 *Among EDGE bases of degree $k \leq 3$ referred to the same null law, the retained non-centrality of a degree-j departure is ρ^{2j} if $j \leq k$ and zero otherwise, while the basis spends k degrees of freedom. Keeping the cubic direction only when $\hat{\rho}^6 \geq \tau$, for a threshold $\tau \in (0, 1)$, therefore retains it exactly when its signal survives at fraction τ. With $\tau = 0.2$ the rule takes $k = 3$ when $\hat{\rho} \geq 0.76$ and $k = 2$ otherwise, and we keep the cubic in addition whenever $\kappa < 0.05$, where there is no selection term to avoid.*

The corollary is elementary, and its value is in the second consequence of Empirical law 1: when the index is noisy the rule drops the cubic direction, which by Proposition 2 retains about $\rho^6 \approx 0.12$ of any cubic departure and by Empirical law 1 carries almost all of the selection non-centrality. One decision removes a powerless direction and a biased one. Two guards are needed, and both exist because $\hat{\rho}$ is a null quantity: under a misspecified model the representation (8) reads the misfit as noise and $\hat{\rho}$ falls, so a rule that used $\hat{\rho}$ alone would discard directions precisely when a departure occupies them. The first guard is the floor at two. A degree-one basis is the calibration-slope direction alone and cannot see any change of shape, and in the simulations of Section 5.5 $\hat{\rho}$ fell from 0.66–0.71 under the null to 0.44–0.60 under the strongest departures, which without a floor would have cost the quadratic direction. The second is the aspect-ratio clause. At fixed p the selection term is negligible, so there is nothing to gain by dropping the cubic, and $\hat{\rho}$ falls from 0.93 under the null to 0.75 under a strong cubic departure — across the threshold, and therefore into discarding the direction that carries the signal, at a cost of 0.60 in power (Section 5.6). Keeping $k = 3$ whenever $\kappa < 0.05$ removes that failure without affecting any proportional-regime cell. What remains is a single decision, whether to keep the cubic when the index is noisy, and the rule makes it correctly on the glaucoma data of Section 6, where $\hat{\rho} = 0.92$ and the misfit lies in the cubic direction. Table 4 reports the sensitivity to τ: values below 0.2 keep the cubic too often and the size rises to 0.61–0.86 in the design where the selection term is largest, while $\tau = 0.3$ behaves like $\tau = 0.2$ and would still keep the cubic on the glaucoma data, where $\hat{\rho}^6 = 0.61$.

We call the result the *adaptive* EDGE basis and write EDGE-ρ where a symbol is needed.

4.5. Definition of CALM

CALM refers the shrinkage-corrected statistics of (4) to the exact weighted chi-squared law of Section 2.3 with covariance $(1 + \hat{g})\Sigma_{\text{dn}}$, where Σ_{dn} is the de-noised reference of Section 4.2, $\hat{g}$ is the basis-specific non-centrality fraction of Empirical law 1 (0.24κ for the decile basis, 1.15κ for EDGE-3 and 0.11κ for EDGE of degree at most two), and the EDGE basis is the adaptive one of Corollary 1 together with the aspect-ratio clause that follows it. It requires one penalised fit, no resampling, and runs in a fraction of a second. At fixed p the inflation is negligible ($\hat{g} = 0.002$) and CALM is the reference (7).

The two bases are not on the same footing in the proportional regime, and Section 5.4 shows why this matters. The adaptive EDGE basis removes the selection term by dropping the direction that carries it, and needs no constant.

Table 1. Rejection rate of a correctly specified model at nominal level 0.05 under four references, decile basis and EDGE basis, eight cells. The first three EDGE columns use EDGE-3; the CALM column uses the adaptive EDGE basis and the basis-specific inflation of Section 4.5. 500 replications per cell.

		Decile basis (SC.HL)				EDGE basis (SC.EDGE)			
Design	λ	first order	Σ_{d}	Σ_{dn}	CALM	first order	Σ_{d}	Σ_{dn}	CALM
A ($\kappa = 0.01$)	100	0.030	0.044	0.046	0.046	0.028	0.050	0.048	0.050
A	137	0.022	0.036	0.040	0.038	0.030	0.040	0.038	0.040
A	200	0.018	0.044	0.046	0.046	0.030	0.050	0.048	0.048
B, $\kappa = 0.10$	416	0.006	0.054	0.066	0.060	0.012	0.048	0.054	0.040
B, $\kappa = 0.25$	50	0.020	0.086	0.046	0.032	0.030	0.120	0.056	0.006
B, $\kappa = 0.25$	416	0.018	0.076	0.072	0.052	0.024	0.080	0.076	0.028
B, $\kappa = 0.25$	1000	0.014	0.072	0.052	0.036	0.034	0.082	0.072	0.030
B, $\kappa = 0.40$	416	0.022	0.128	0.054	0.036	0.036	0.108	0.084	0.018

The decile basis spends all G directions, so it cannot avoid that direction and must instead absorb it into the reference through $\hat{g}$, a quantity that was fitted on eight cells. We therefore recommend the adaptive EDGE basis as the primary statistic and report the decile statistic, which is the shrinkage-corrected Hosmer–Lemeshow test and the one a reader of that tradition will look for, alongside it.

5. Numerical results

5.1. Designs

We use the registered designs of Ebrahim (2026) so that every number can be compared with that paper. Design A has $n = 500$, $p = 5$, independent standard normal covariates, $\boldsymbol{\beta} = (0.5, -0.4, 0.3, -0.3, 0.2)^\top$ and $\lambda \in \{100, 137, 200\}$. Design B has $n = 400$, an autoregressive covariate correlation of 0.7, coefficients cycling $(0.35, -0.30, 0.25, -0.20, 0.15)$ scaled to $\|\boldsymbol{\beta}\| = 2.31$, and $p = 100$ ($\kappa = 0.25$) with $\lambda \in \{50, 416, 1000\}$; a κ sweep keeps $\lambda = 416$ and takes $p = 40$ ($\kappa = 0.10$) and $p = 160$ ($\kappa = 0.40$). All fits are ridge logistic regression by penalised iteratively reweighted least squares with the intercept unpenalised; $G = 10$ throughout. Under the null, data are generated from the design's own model, so every rejection is a false one. Sizes are based on 500 replications unless stated, for a Monte Carlo standard error of 0.010 at the 5 per cent level.

5.2. Size

Table 1 and Figure 3 give the size of the four references. The first-order reference is conservative everywhere and severely so at $\kappa = 0.10$, where it rejects 0.6 per cent of correct models. The variance at the debiased fit is exact at fixed p and at $\kappa = 0.10$ and liberal at $\kappa \geq 0.25$. The de-noised variance repairs the lightest-penalty cell and halves the excess at $\kappa = 0.40$. CALM is within Monte Carlo error of the nominal level in every cell on both bases and, where it departs from it, does so towards conservatism: on the EDGE basis in the proportional regime the rule of Corollary 1 drops the cubic direction, and the two-dimensional basis rejects 0.6 to 4 per cent of correct models, least at $\kappa = 0.25$, $\lambda = 50$. That conservatism is not caused by the inflation. On the EDGE basis the factor is at most 1.05, and removing it altogether changes the size by at most 0.008 in every cell we examined; the de-noised reference overstates the null variance in the low-degree directions while understating it in the cubic direction, and dropping the cubic leaves only the overstated part. We keep the inflation on this basis for consistency of definition rather than for its effect, and the statistic would behave the same without it.

5.3. Robustness to the design law

Assumption 2 requires Gaussian predictors. Table 2 repeats three cells with predictors drawn from a t_3 distribution scaled to unit variance and from a standardised Bernoulli(0.3) law, the correlation structure imposed through the same Cholesky factor. At fixed p the size of the de-noised reference is 0.042–0.060 under all three laws and the estimated scale of the linear predictor is recovered under each; at $\kappa > 0$ an excess of one to five points appears

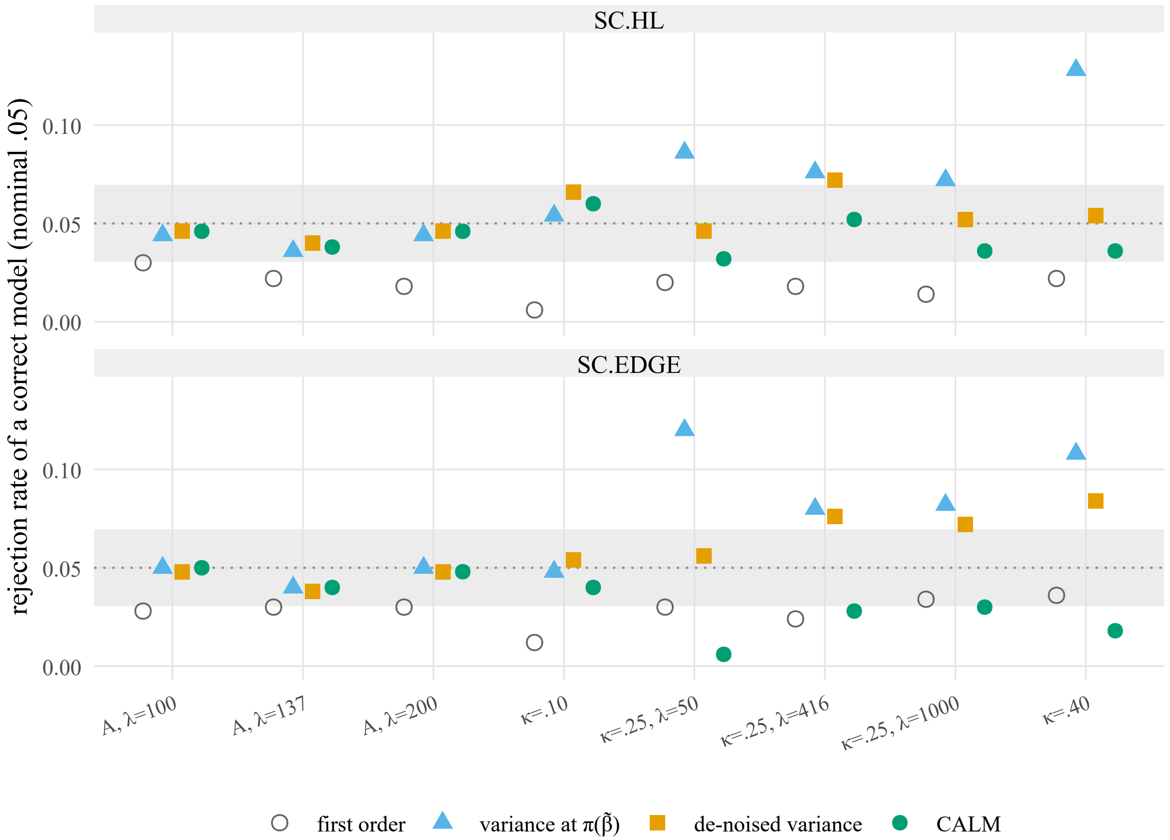

Figure 3 Size of the four references across the eight cells at nominal level 0.05, decile basis above and EDGE basis below. The band is two Monte Carlo standard errors.

Table 2. Size at nominal level 0.05 of the de-noised reference Σ_{dn} under three predictor laws, decile and EDGE-3 bases, with the estimated standard deviation of the linear predictor (true values 0.79 in design A and 1.48–1.50 in design B). 400 replications per cell.

Design	law	decile	EDGE-3	estimated s.d.
A	Gaussian	0.048	0.042	0.79
A	t_3	0.052	0.052	0.77
A	binary	0.058	0.060	0.80
B, $\kappa = 0.10$	Gaussian	0.060	0.045	1.45
B, $\kappa = 0.10$	t_3	0.062	0.098	1.38
B, $\kappa = 0.10$	binary	0.070	0.058	1.44
B, $\kappa = 0.25$	Gaussian	0.062	0.077	1.43
B, $\kappa = 0.25$	t_3	0.055	0.065	1.37
B, $\kappa = 0.25$	binary	0.080	0.052	1.43

under every law before the adaptive basis and inflation are applied, largest for the EDGE-3 basis under t_3 predictors at $\kappa = 0.10$. Section 5.4 runs the whole procedure under these laws and finds the same picture as for the Gaussian designs, so the Gaussian assumption is a condition of the proof rather than the boundary of the method.

Table 3. Rejection rate of a correctly specified model at nominal level 0.05 in designs that were not used to calibrate the constants of Empirical law 1: nine designs with the Gaussian predictor law and eight repetitions of earlier cells under t_3 and standardised Bernoulli predictors. The last column is the mean of the observable $\hat{\rho}$. 500 replications per cell; the Monte Carlo standard error is 0.010.

Design	SC.HL	adaptive EDGE	EDGE-3	$\hat{\rho}$
New designs, Gaussian predictors				
$\kappa = 0.15$ $(p = 60)$	0.066	0.054	0.080	0.73
$\kappa = 0.33$ $(p = 132)$	0.052	0.030	0.060	0.72
$\kappa = 0.40$, $\lambda = 50$	0.130	0.024	0.138	0.70
$\kappa = 0.40$, $\lambda = 1000$	0.034	0.056	0.062	0.73
$\kappa = 0.25$, correlation 0.3	0.688	0.076	0.872	0.70
$\kappa = 0.25$, $\lVert\boldsymbol{\beta}\rVert = 1.5$	0.054	0.028	0.030	0.69
$\kappa = 0.25$, $\lVert\boldsymbol{\beta}\rVert = 3.0$	0.082	0.080	0.132	0.74
$\kappa = 0.25$, $n = 800$, $p = 200$	0.122	0.074	0.168	0.74
$\kappa = 0.25$, $G = 20$	0.044	0.056	0.102	0.73
Other predictor laws				
A, t_3	0.050	0.050	0.050	0.92
A, binary	0.052	0.058	0.058	0.94
$\kappa = 0.10$, t_3	0.112	0.080	0.164	0.72
$\kappa = 0.10$, binary	0.064	0.074	0.094	0.73
$\kappa = 0.25$, t_3	0.066	0.044	0.084	0.72
$\kappa = 0.25$, binary	0.054	0.044	0.072	0.72
$\kappa = 0.40$, t_3	0.076	0.050	0.094	0.71
$\kappa = 0.40$, binary	0.056	0.044	0.078	0.72

5.4. Validation outside the calibration cells

The constants of Empirical law 1 were fitted on the eight cells of Table 1, and Table 1 therefore reports the size of CALM in the cells that calibrated it. Table 3 reports it in seventeen cells that did not: two intermediate aspect ratios, two penalties at $\kappa = 0.40$, a design with covariate correlation 0.3 instead of 0.7, a weaker and a stronger signal, a design twice the size at the same aspect ratio, twenty groups instead of ten, and the three predictor laws of Section 5.3 run through the whole procedure rather than through Σ_{dn} alone.

The two bases behave differently, and the difference is the main practical finding of this section. On the adaptive EDGE basis the size lies between 0.024 and 0.080 in every one of the seventeen cells, with fifteen of the seventeen within three Monte Carlo standard errors of the nominal level and the largest at 0.080; the basis needs no calibrated constant, because the rule of Corollary 1 drops the direction the selection term occupies rather than correcting for it. On the decile basis, which spends all G directions and must therefore absorb that term into the reference through $\hat{g}$, the size is between 0.034 and 0.076 in twelve cells and fails in the remaining five, most severely at covariate correlation 0.3, where it rejects 0.688 of correct models.

The failures have one cause, and it is visible in Empirical law 1 itself. The selection term is not a function of κ. An observation changes group when its own response moves its fitted index across a group boundary, so the term grows with the self-influence $\bar{h}$ measured against the spread of the fitted index, and with the number of observations per group. At $\kappa = 0.25$ the squared mean $\lVert\mathbb{E}\boldsymbol{v}\rVert^2$ is seven times larger under covariate correlation 0.3 than under 0.7, because the fitted index has half the spread there while the self-influence is unchanged, and it doubles when n doubles at fixed κ. The quantity $n\bar{h}^2/\mathrm{sd}(\hat{\eta})^2$, which the boundary-crossing argument suggests, tracks the term across these designs better than κ does but not closely enough to replace it, and a per-dataset estimate built from the leave-one-out displacement of the index proved far too noisy to use. A correction of the decile statistic that is valid across design families therefore remains open, and until it exists the decile reference should be read as calibrated for designs resembling those of Table 1. The prepivoting bootstrap is not a way out: in the two failing designs we also ran it, its size is 0.11 and 0.07, so the difficulty is in the corrected statistic and not in the analytic reference alone.

Table 4. Sensitivity of the size of the adaptive EDGE statistic to the threshold τ of Corollary 1, at nominal level 0.05, in seven designs. The value used throughout is $\tau = 0.2$. 500 replications per cell.

	threshold τ			
Design	0.05	0.10	0.20	0.30
A ($\kappa = 0.01$)	0.050	0.050	0.050	0.050
B, $\kappa = 0.10$	0.070	0.070	0.052	0.040
B, $\kappa = 0.25$	0.062	0.064	0.044	0.032
B, $\kappa = 0.40$	0.048	0.040	0.032	0.024
$\kappa = 0.40$, $\lambda = 50$	0.138	0.092	0.024	0.022
$\kappa = 0.25$, correlation 0.3	0.856	0.610	0.076	0.024
$\kappa = 0.10$, t_3	0.162	0.156	0.080	0.048

5.5. Power

An exact reference has no Monte Carlo error in its p-value, and a bootstrap with B refits does; one may therefore hope for a small gain in power at no cost in size. But the more useful question is how CALM compares with what an analyst would otherwise use, and Figure 4 and Table 5 answer it on identical simulated datasets in the four κ cells, against two departures a grouped test should be able to see: a Hermite quadratic along the true index and an omitted interaction $x_1 x_2$ off the index, each at 0.5, 1 and 1.5 times the standard deviation of the linear predictor for the first and 1 and 1.5 times for the second. The comparison set is the prepivoting bootstrap of Ebrahim (2026) with 399 refits, the generalised residual prediction (GRP) test of Janková et al. (2020) with its own lasso fit (R package `GRPtests` 0.1.2, default settings: five sample splits, lasso tuned by cross-validation), the classical Hosmer–Lemeshow test on the unpenalised maximum likelihood fit where that fit exists, and the four checks a clinical prediction paper would report on the ridge fit itself: the Hosmer–Lemeshow test without correction, the calibration intercept-and-slope test (Cox, 1958), Spiegelhalter's z (Spiegelhalter, 1986) and the unweighted sum of squares of Copas (1989), together with the Hosmer–Lemeshow test on ten-fold cross-validated predictions. A test's power curve is drawn only in cells where it holds its level; a rejection rate above a broken level is not power, and those tests appear in Table 6 with their size and their size-adjusted power, the rejection rate at the critical value that would give the test its nominal size in that cell. Every check a clinical prediction paper would report on the ridge fit rejects almost every correctly specified model, at fixed p as well as in the proportional regime, because each of them is displaced by shrinkage in the way Section 2 describes for the Hosmer–Lemeshow statistic; and even size-adjusted, the calibration-slope and Spiegelhalter statistics do not rank the departures above the null, because a quadratic departure moves the apparent slope back towards one; the uncorrected Hosmer–Lemeshow statistic keeps its ranking power at fixed p (0.84) and loses it in the proportional regime. The test on cross-validated predictions rejects for a different reason: shrunk predictions are miscalibrated on new data even when the model is correct, so it tests the calibration of the deployed probabilities and not the correctness of the model.

Three things are visible. First, against the bootstrap CALM loses no power. On the EDGE basis the paired differences against the quadratic departure are within ± 0.04 at every κ (0.640 against 0.632 at fixed p and relative size 0.5; 0.708 against 0.695 and 0.895 against 0.875 at $\kappa = 0.10$; 0.420 against 0.458 and 0.640 against 0.615 at $\kappa = 0.25$; 0.260 against 0.295 and 0.350 against 0.367 at $\kappa = 0.40$), with 399 refits replaced by one fit. On the decile basis CALM matches the bootstrap at fixed p and at $\kappa = 0.10$ and gives up 0.10–0.12 at $\kappa = 0.25$ (0.233 against 0.333, 0.310 against 0.432); about half of that gap is the inflation $(1 + 0.24\kappa)$ that protects its level and half is the de-noised variance itself, since the un-inflated reference Σ_{dn} gives 0.280 and 0.368 on the same data; at $\kappa = 0.40$ the bootstrap's higher rates (0.237, 0.302) sit on a size of 0.074. Second, CALM recovers the classical power. At fixed p the Hosmer–Lemeshow test on the maximum likelihood fit rejects the quadratic departure at 0.390, 0.983 and 0.997; CALM on the ridge fit, after subtracting the displacement and correcting the reference, rejects it at 0.343, 0.965 and 0.995, and its EDGE basis at 0.640, 0.998 and 1. Third, CALM and the test of Janková et al. (2020) see different things, as Ebrahim (2026) found on real data. The generalised residual prediction test is the stronger test off the index — 0.457 and 0.897 against the interaction at fixed p, where CALM's EDGE basis gives 0.352 and 0.517 and no grouped test has power once $\kappa > 0$ — and the weaker along it, with 0.020, 0.407 and 0.877 against the quadratic at fixed p and 0.010, 0.083 and 0.297 at $\kappa = 0.10$, where CALM gives 0.233, 0.708 and 0.895. It also keeps some of its off-index power in the proportional regime (0.247–0.420 at $\kappa = 0.25$, 0.253–0.340

Table 5. Power at nominal level 0.05 of the tests that hold their level, on identical datasets, against a quadratic departure along the index and an omitted interaction off the index. Rows with relative size 0 are size. Entries are omitted where the maximum likelihood estimate did not exist in most datasets or where the test does not hold its level in that cell. 300–500 replications per cell.

			CALM		prepivot			MLE fit	
Design	departure	size	SC.HL	SC.EDGE	SC.HL	SC.EDGE	GRP	HL	EDGE
A ($\kappa = 0.01$)	—	0	0.050	0.043	0.044	0.046	0.003	0.067	0.053
	quadratic in the index	0.5	0.343	0.640	0.328	0.632	0.020	0.390	0.650
	quadratic in the index	1	0.965	0.998	0.950	0.998	0.407	0.983	0.997
	quadratic in the index	1.5	0.995	1.000	0.995	1.000	0.877	0.997	1.000
	x_1x_2 off the index	1	0.215	0.352	0.200	0.350	0.457	0.267	0.413
	x_1x_2 off the index	1.5	0.347	0.517	0.338	0.537	0.897	0.357	0.560
B, $\kappa = 0.10$	—	0	0.052	0.033	0.050	0.042	0.007	0.097	0.067
	quadratic in the index	0.5	0.098	0.233	0.080	0.217	0.010	—	—
	quadratic in the index	1	0.405	0.708	0.440	0.695	0.083	—	—
	quadratic in the index	1.5	0.695	0.895	0.713	0.875	0.297	—	—
	x_1x_2 off the index	1	0.028	0.022	0.028	0.033	0.557	—	—
	x_1x_2 off the index	1.5	0.048	0.020	0.045	0.035	0.807	—	—
B, $\kappa = 0.25$	—	0	0.040	0.030	0.058	0.046	0.007	0.326	0.237
	quadratic in the index	0.5	0.080	0.140	0.095	0.152	0.020	—	—
	quadratic in the index	1	0.233	0.420	0.333	0.458	0.067	—	—
	quadratic in the index	1.5	0.310	0.640	0.432	0.615	0.127	—	—
	x_1x_2 off the index	1	0.040	0.048	0.052	0.070	0.247	—	—
	x_1x_2 off the index	1.5	0.045	0.037	0.065	0.062	0.420	—	—
B, $\kappa = 0.40$	—	0	0.035	0.018	0.074	0.050	0.043	—	—
	quadratic in the index	0.5	0.068	0.077	0.145	0.125	0.057	—	—
	quadratic in the index	1	0.138	0.260	0.237	0.295	0.090	—	—
	quadratic in the index	1.5	0.168	0.350	0.302	0.367	0.070	—	—
	x_1x_2 off the index	1	0.048	0.035	0.100	0.065	0.253	—	—
	x_1x_2 off the index	1.5	0.040	0.043	0.098	0.062	0.340	—	—

at $\kappa = 0.40$), where the interaction is absorbed by the ridge fit and no grouped test sees it. One test asks whether the model is right as a function of the covariates, the other whether the deployed index is calibrated along itself; an analyst who wants both should run both.

Designs A and B are dense, while the test of Janková et al. (2020) assumes a sparse coefficient vector, so the comparison above is made outside its own assumptions and its conservatism there (0.003 to 0.043 under the null) may be a consequence of that. We therefore repeated the comparison at $\kappa = 0.25$ with only five non-zero coefficients out of 100, where it is on its own ground. Its size is then 0 in 300 replications, still conservative; against the off-index interaction it rejects 0.87, against 0.03 for CALM, and against the quadratic along the index it rejects 0.077 against 0.21 for the adaptive EDGE basis. Sparsity therefore sharpens the division of labour rather than changing it: the covariate-function test finds what is off the index, the calibration test finds what is along it. The classical test on the maximum likelihood fit, finally, is not an option in the proportional regime: it rejects 9.7 per cent of correct models at $\kappa = 0.10$ and 33 per cent at $\kappa = 0.25$, and at $\kappa = 0.40$ the maximum likelihood estimate did not exist in any of the 300 datasets (Candès and Sur, 2020).

5.6. The basis degree and the attenuation law

Figure 5 tests Corollary 1 directly. Against a degree-two Hermite departure along the index, the degree-matched basis EDGE-2 has more power than EDGE-3 in every cell and the difference grows with κ — 0.510 against 0.492 at $\kappa = 0.25$ and rel $= 1$, 0.704 against 0.650 at rel $= 1.5$ — while both leave the decile basis far behind (0.360). Against a degree-three departure, only EDGE-3 has power at fixed p (0.558), and at $\kappa \geq 0.10$ no basis has any (0.04–0.09), as $\hat{\rho}^6$ predicts: it is 0.12 under the null in those cells and falls to 0.06 under the departure itself. The

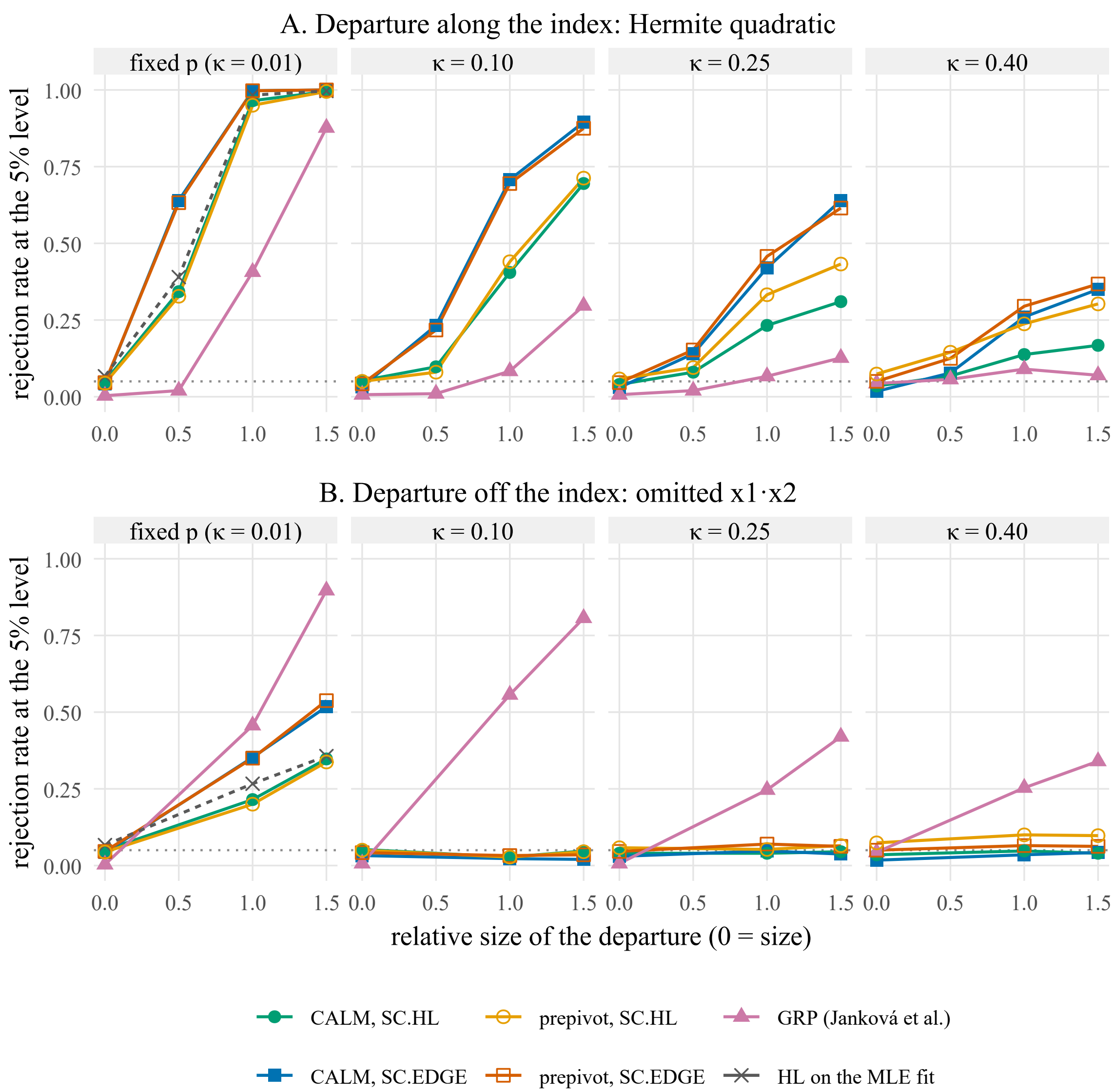

Figure 4 Power at nominal level 0.05 against a quadratic departure along the index (A) and an omitted interaction off the index (B), by relative size of the departure, in the four κ cells, on identical datasets: CALM on both bases, prepivoting with 399 refits, the generalised residual prediction test and the Hosmer–Lemeshow test on the maximum likelihood fit. A curve is drawn only where the test holds its level in that cell; the tests that hold it nowhere are in Table 6. Relative size 0 is size. 300–500 replications per point.

observable $\hat{\rho}$ is 0.94, 0.71 and 0.69 in the three cells under the null; the rule of Corollary 1 selects degree three, two and two, which is the most powerful choice in every row of the figure.

The aspect-ratio clause of the rule can be seen at work in a separate run of 500 replications in the same three cells. Against a cubic departure at fixed p, $\hat{\rho}$ falls from 0.93 under the null to 0.84 at relative size 1 and 0.70 at relative size 1.5. A rule that used $\hat{\rho}$ alone would then keep the cubic in only 88 and 44 per cent of replications and reject at 0.496 and 0.372; keeping $k = 3$ whenever $\kappa < 0.05$ raises those to 0.542 and 0.666, matching the fixed

Table 6. Calibration checks applied to the ridge fit: rejection rate of a correctly specified model (size) and size-adjusted power against the quadratic departure of relative size 1, where the critical value is the empirical 5 per cent point of the same test's null distribution in the same cell. A dash marks cells in which every null p-value was below 10^{-10}, so that no empirical critical value exists.

	A		$\kappa = 0.10$		$\kappa = 0.25$		$\kappa = 0.40$	
Test	size	adj. power	size	adj. power	size	adj. power	size	adj. power
uncorrected Hosmer–Lemeshow	0.933	0.840	1.000	0.093	1.000	—	1.000	—
calibration intercept and slope	1.000	0.000	1.000	0.000	1.000	—	1.000	—
Spiegelhalter's z	1.000	0.003	1.000	0.000	1.000	0.000	1.000	—
unweighted sum of squares	1.000	—	1.000	—	1.000	—	1.000	—
Hosmer–Lemeshow on cross-validated predictions	0.593	0.873	0.717	0.140	0.420	0.080	0.250	0.077
Hosmer–Lemeshow on the MLE fit	0.067	0.970	0.097	0.820	0.326	0.197	—	—

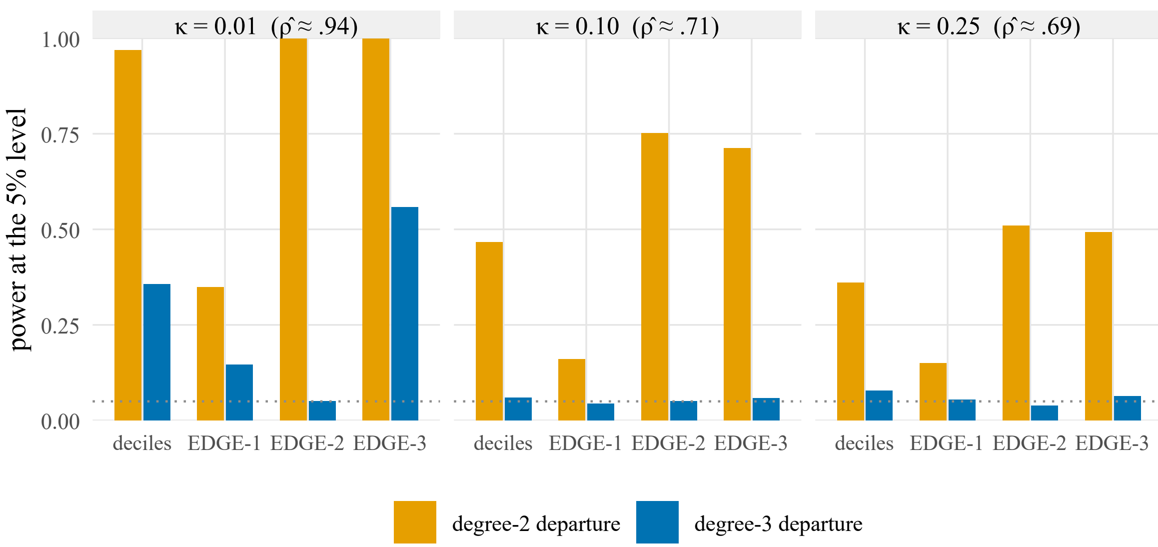

Figure 5 Power at nominal level 0.05 of the decile basis and of EDGE bases of degree one to three against degree-two and degree-three Hermite departures along the index of size rel $= 1$, at $\kappa = 0.01$, 0.10 and 0.25. 500 replications per cell.

EDGE-3 basis exactly. In the two proportional-regime cells the two rules agree in every replication, so the clause costs nothing there. Against a quadratic departure the adaptive basis rejects at 0.916 and 0.980 at $\kappa = 0.10$ and at 0.690 and 0.846 at $\kappa = 0.25$, against 0.642 and 0.912, and 0.426 and 0.542, for the decile basis, which is the attenuation law of Proposition 2 read as a statement about power.

5.7. The boundary: departures that do not survive attenuation

Table 7 shows what the attenuation law forbids, and it is the reason the departures of Section 5.5 were chosen as they were. Against a cubic departure along the index and a small interaction off it, CALM and the bootstrap both hold their level at every κ, and neither has power once $\kappa > 0$: every rate in the proportional regime is within Monte Carlo error of the corresponding size, and the bootstrap's 0.098 and 0.104 at $\kappa = 0.40$ sit on a size of 0.074. This is Proposition 2 and not a property of either reference. With $\hat{\rho} \approx 0.7$ a cubic departure retains about $\rho^6 \approx 0.12$ of its non-centrality, so no grouped test on the fitted index can see it, and the interaction is absorbed by the ridge fit. At fixed p, where $\hat{\rho} \approx 0.9$, both tests see the cubic (0.304 and 0.322 on the EDGE basis at relative size 0.7) and CALM is again level with the bootstrap. What CALM offers in the proportional regime is therefore validity, with power against the departures that survive the index.

Table 7. Size and rejection rates at nominal level 0.05 of CALM and of prepivoting with 399 refits, paired on identical datasets, against departures that do not survive attenuation — a Hermite cubic along the true index and a small omitted interaction x_1x_2, each at relative size 0.4 and 0.7 of the linear predictor. Rows with relative size 0 are size. k is the EDGE degree chosen by Corollary 1 in the majority of replications; diff is the paired difference CALM minus prepivot with its standard error. 500 replications per cell.

				Decile basis (SC.HL)			Adaptive EDGE basis		
Design	departure	size	k	CALM	prepivot	diff (s.e.)	CALM	prepivot	diff (s.e.)
A ($\kappa = 0.01$)	—	0	3	0.046	0.044	$+0.002$ (0.005)	0.050	0.046	$+0.004$ (0.005)
	cubic in the index	0.4	3	0.108	0.106	$+0.002$ (0.004)	0.116	0.118	-0.002 (0.006)
	cubic in the index	0.7	3	0.194	0.184	$+0.010$ (0.007)	0.304	0.322	-0.018 (0.008)
	x_1x_2 off the index	0.4	3	0.082	0.078	$+0.004$ (0.004)	0.112	0.108	$+0.004$ (0.007)
	x_1x_2 off the index	0.7	3	0.144	0.124	$+0.020$ (0.007)	0.250	0.258	-0.008 (0.009)
B, $\kappa = 0.10$	—	0	2	0.060	0.050	$+0.010$ (0.005)	0.040	0.042	-0.002 (0.009)
	cubic in the index	0.4	2	0.036	0.040	-0.004 (0.005)	0.020	0.022	-0.002 (0.007)
	cubic in the index	0.7	2	0.026	0.030	-0.004 (0.003)	0.020	0.040	-0.020 (0.008)
	x_1x_2 off the index	0.4	2	0.046	0.034	$+0.012$ (0.006)	0.032	0.044	-0.012 (0.007)
	x_1x_2 off the index	0.7	2	0.050	0.048	$+0.002$ (0.003)	0.044	0.052	-0.008 (0.011)
B, $\kappa = 0.25$	—	0	2	0.052	0.058	-0.006 (0.005)	0.028	0.046	-0.018 (0.010)
	cubic in the index	0.4	2	0.038	0.062	-0.024 (0.007)	0.026	0.036	-0.010 (0.007)
	cubic in the index	0.7	2	0.036	0.078	-0.042 (0.009)	0.026	0.032	-0.006 (0.008)
	x_1x_2 off the index	0.4	2	0.034	0.044	-0.010 (0.004)	0.028	0.044	-0.016 (0.010)
	x_1x_2 off the index	0.7	2	0.042	0.062	-0.020 (0.006)	0.022	0.058	-0.036 (0.011)
B, $\kappa = 0.40$	—	0	2	0.036	0.074	-0.038 (0.009)	0.018	0.050	-0.032 (0.009)
	cubic in the index	0.4	2	0.042	0.098	-0.056 (0.010)	0.012	0.050	-0.038 (0.009)
	cubic in the index	0.7	2	0.026	0.104	-0.078 (0.012)	0.020	0.048	-0.028 (0.010)
	x_1x_2 off the index	0.4	2	0.046	0.076	-0.030 (0.008)	0.024	0.062	-0.038 (0.010)
	x_1x_2 off the index	0.7	2	0.058	0.086	-0.028 (0.008)	0.032	0.062	-0.030 (0.011)

5.8. Other smooth penalties

Nothing in Sections 2–4 uses the linearity of the ridge penalty. If a smooth penalty contributes a score term $\boldsymbol{a}(\boldsymbol{\beta})$ with $K(\boldsymbol{\beta}) = -\partial \boldsymbol{a}/\partial \boldsymbol{\beta}$, then $\tilde{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}} - F^{-1}\boldsymbol{a}(\hat{\boldsymbol{\beta}})$, $\hat{\boldsymbol{\mu}} = -UM^{-1}\boldsymbol{a}(\tilde{\boldsymbol{\beta}})$ and Theorem 1 go through with K evaluated at $\hat{\boldsymbol{\beta}}$ (Ebrahim, 2026, Remark 2). Section S1 of the supplementary material reports the generalised ridge penalty and Firth's penalty (Firth, 1993), the standard response to separation in clinical prediction (Puhr et al., 2017), whose implementation we checked against the extttlogistf package (Heinze and Schemper, 2002). The first behaves exactly as the ridge penalty does. The second marks a boundary of the framework: at fixed p Firth's correction is $O(n^{-1})$ and there is nothing to correct, but in the proportional regime it is not a shrinkage penalty at all, the fitted coefficient norm exceeding the true one, so the displacement that this paper removes has the wrong sign and the corrected references reject 12 to 32 per cent of correct models. The framework is for estimators that shrink towards zero.

6. The glaucoma model revisited

Ebrahim (2026) analysed a ridge logistic model for glaucoma diagnosis from $p = 62$ morphometric measurements of the optic nerve head on $n = 196$ eyes (the GlaucomaM data; see Coan et al., 2023 for the clinical background), a design in which the maximum likelihood estimate does not exist (Candès and Sur, 2020) and $\kappa = 0.32$. At the cross-validated penalty the uncorrected Hosmer–Lemeshow test returned $p < 10^{-5}$; the prepivoted corrected test returned 0.026 on the decile basis and 0.034 on the EDGE basis, and the paper concluded that the evidence for misfit was real but modest and that the model needed recalibration rather than rebuilding. Table 8 adds the analytic references. The first-order reference is conservative, as Theorem 1 predicts at this κ; the variance at the debiased fit gives 0.024 and 0.034, within 0.002 of the bootstrap; CALM gives 0.012 and 0.033, with $\hat{\rho} = 0.92$ so that the adaptive rule keeps all three EDGE directions. The de-noised reference is the most liberal of the four, at 0.007 and 0.011, and the inflation $1 + 0.24\kappa = 1.08$ brings the decile p-value back to 0.012. De-noising lowers the p-value here because the signal is strong and the index is accurate: the observables are $\hat{a} = 1.00$, $\hat{\sigma} = 0.43$ and $\hat{c} = 3.14$,

Table 8. The glaucoma model at the cross-validated penalty $\lambda = 295$ ($\lambda_{\mathrm{g}} = 1.51$), $G = 10$: p-values of the uncorrected test and of the corrected statistics under four references.

Reference	SC.HL	SC.EDGE
uncorrected statistic, maximum likelihood covariance	$< 10^{-5}$	$< 10^{-5}$
first order (Ω_{MLE})	0.163	0.128
variance at the debiased fit (Σ_{d})	0.024	0.034
de-noised variance (Σ_{dn})	0.007	0.011
CALM	0.012	0.033
prepivoted, 499 refits (Ebrahim, 2026)	0.026	0.034

giving an estimated standard deviation of 3.15 for the true linear predictor against 1.11 for the fitted one, so the estimated null Bernoulli variance is well below the fitted variance and $\mathrm{tr}(\Sigma_{\mathrm{dn}})$ is 0.82 of $\mathrm{tr}(\Sigma_{\mathrm{d}})$. Two cautions belong with the number. This dataset lies outside every simulated design: $\hat{\rho} = 0.92$ against 0.65 to 0.74 in the proportional-regime cells on which the constants were fitted, and the maximum likelihood estimate does not exist, so the scale estimate operates in a regime the simulations did not probe. And the misfit is in the cubic direction: the three EDGE coordinates of $\boldsymbol{v}$ are -0.16, 0.14 and -1.38, so a degree-two basis would return 0.57 rather than 0.033. The computation takes under half a second against 3.3 seconds for prepivoting with 499 refits, and its p-value carries no Monte Carlo error. The conclusion of the earlier paper stands, and no longer depends on a bootstrap.

7. Discussion

The general lesson is Theorem 1. Every residual diagnostic for a penalised generalised linear model that standardises by the fitted variance inherits an error of order λ/n in the direction of conservatism, because shrinkage moves fitted values towards the region where the variance function is largest. The grouped tests treated here are the case that clinical prediction practice uses, but the mechanism is not specific to them, and we would expect Pearson-type and deviance-type statistics for penalised fits to show it as well. Section 5.5 shows how far the lesson reaches in practice: every calibration check in clinical use — the uncorrected Hosmer–Lemeshow test, the calibration slope, Spiegelhalter's z, the unweighted sum of squares — rejects almost every correctly specified penalised model, and the one high-dimensional test that does hold its level, that of Janková et al. (2020), answers a different question and is blind along the index where CALM sees. The two are complements, not competitors.

What is proved and what is calibrated should be kept apart. Theorem 1 and Corollary 1 are proved. Proposition 1 rests on the theorems of Bellec (2025) under a Gaussian design and on the consistency of a one-dimensional scale estimate that we assume rather than prove. Empirical law 1 is an empirical law with constants fitted on eight cells from two design families; we present the inflation it defines as a theory-motivated correction of order κ whose constant is calibrated, and we note that the adaptive basis, which needs no constant at all, holds the level at every κ examined on its own. The order of the selection term follows from the self-influence argument of Section 4.3, but a derivation of its constant from the mean-field theory is open; the mean-field prediction of the per-group means reproduces their shape with correlation 0.97 and their scale to within 27 per cent, which explains the term and does not yet subtract it.

Four boundaries are stated plainly. The first is the one Section 5.4 draws between the two bases: the adaptive EDGE statistic held its level in every design we tried, while the decile statistic depends on a constant fitted on eight cells and fails where the fitted index has little spread relative to the self-influence, so a correction of the decile statistic that transfers across design families is open. The second is the regime: the reference is validated for $\kappa \leq 0.4$ and for penalties that shrink towards zero, the theory is for Gaussian designs with robustness beyond them empirical, and bias-reduced estimators in the proportional regime fall outside it altogether (Section 5.8). The third is that the penalty λ is treated as fixed, as in Ebrahim (2026); the selection effect of choosing it by cross-validation on the same data is not accounted for here, nor, so far as we know, by any test in this class. The fourth is power: the power of any grouped test on a noisy index is limited by Proposition 2, so that at $\kappa \geq 0.25$ what CALM offers is validity with power against the departures that survive the index, not power against all.

What should an analyst do with this? Fit the model, choose the penalty as usual, and run CALM on the adaptive EDGE basis; that is the statistic we would report, and the decile version alongside it if a reader expects the

Hosmer–Lemeshow form. Read a rejection as evidence that the logistic form is wrong along the fitted index, and a non-rejection as saying nothing about whether the shrunk probabilities are calibrated, which under a penalty they are not. The method covers ridge, generalised ridge and any other smooth penalty that shrinks towards zero; it does not cover the lasso, whose penalty is not differentiable, and the elastic net only through its ridge part.

A practical remark closes the paper. Grouping the observations on an index that depends less on the response — the fitted index of a much heavier ridge fit, or a principal component of the design — also removes the selection term, and in simulation gives a valid test without any calibrated constant. We do not recommend it, and the glaucoma data say why: forming the deciles from a fit at five times the penalty re-ranks 38 per cent of the patients and the evidence for misfit disappears ($p = 0.63$). The misfit in that model is a failure of calibration along the deployed index specifically. A test that groups on a different index tests a different hypothesis. CALM corrects the reference and leaves the question alone.

Data availability

The `GlaucomaM` data are distributed in the R package `TH.data` on the Comprehensive R Archive Network. All code needed to reproduce every number, table and figure in this paper, with random seeds and session information, is provided in the reproduction archive that accompanies the manuscript as supplementary material and is deposited at `https://doi.org/10.5281/zenodo.22467285`. The archive contains a reference implementation, `calm.gof()`, which will be added to the R package `ebrahim.gof`; the prepivoting comparator is `shrink.gof()` in version 2.6.0 of that package, on the Comprehensive R Archive Network.

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Conflict of interest

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